1 Schubert et al., 2001 General equations

This document/section has a different approach to structure the derivation of the ALA: this is a summary of the derivation of the ALA done by Schubert et al., 2001. red text means assumtions/approximations were made toward the ALA.

1.1 Mass equation/continuity equation/conservation of mass

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x_i}(\rho u_i) = \frac{1}{\rho}\frac{D\rho}{Dt} + \frac{\partial u_i}{\partial x_i} = 0 \tag{1}$$

where $\frac{D}{Dt}$ represents the material derivative ((**??**)). For incompressible flow, $\frac{D\rho}{Dt} = 0$, so Eq. (1) reduces to:

$$\frac{\partial u_i}{\partial x_i} = \nabla \cdot \mathbf{u} = 0 \tag{2}$$

For the special case of an incompressible fluid, $\frac{D\rho}{Dt} = 0$, so Eq. (2) reduces to

$$\frac{\partial u_i}{\partial x_i} = 0 \tag{3}$$

1.2 Momentum equations/conservation of momentum

Starting governing equation (6.4.1):

$$\rho\frac{Du_i}{Dt} = -\frac{\partial p}{\partial x_i} + \frac{\partial \tau_{ij}}{\partial x_j} + \rho g_i \tag{4}$$

we assume gravity g to be the only body force.The coriolis force is negligibly small in the Earth's mantle, and the centrifugal force due to the Earth's rotation is included in the gravity term.

In 2D, the momementum equation components are:

$$\rho\left(\frac{\partial u_x}{\partial t} + u_x\frac{\partial u_x}{\partial x} + u_y\frac{\partial u_x}{\partial y}\right) = -\frac{\partial p}{\partial x} + \frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{xy}}{\partial y}, \tag{5}$$

$$\rho\left(\frac{\partial u_y}{\partial t} + u_x\frac{\partial u_y}{\partial x} + u_y\frac{\partial u_y}{\partial y}\right) = -\frac{\partial p}{\partial y} + \frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \tau_{yy}}{\partial y} + \rho g. \tag{6}$$

Note that the assumption is made that gravity acts in the positive y-direction.

1.3 Navier-Stokes equation

The local strain rate tensor e_{ij} is given by:

$$e_{ij} = \frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right) \tag{7}$$

For a Newtonian, isotropic fluid, the relation between the deviatoric(!) stress tensor and the strain rate tensor is

$$\tau_{ij} = 2\mu e_{ij} + \lambda e_{kk}\delta_{ij} \tag{8}$$

Defining the bulk viscosity k_B:

$$\frac{\tau_{ii}}{3} = e_{ii}\left(\lambda + \frac{2}{3}\mu\right) = k_B e_{ii} \tag{9}$$

The combination of Eq. (8), (9) and (7) gives

$$\tau_{ij} = \mu\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right) + \left(k_B - \frac{2}{3}\mu\right)\frac{\partial u_k}{\partial x_k}\delta_{ij} \tag{10}$$

and substitution of Eq. (10) into Eq. 4 gives

$$\rho\frac{Du_i}{Dt} = -\frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j}\left[\mu\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right) + \left(k_B - \frac{2}{3}\mu\right)\frac{\partial u_k}{\partial x_k}\delta_{ij}\right] + \rho g_i \tag{11}$$

For many fluids, k_B is taken to be zero **(Stokes assumption)**, so Eq. (10) becomes

$$\tau_{ij} = \mu\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} - \frac{2}{3}\frac{\partial u_k}{\partial x_k}\delta_{ij}\right) \tag{12}$$

and Eq. (11) becomes the Navier-Stokes equation:

$$\rho\frac{Du_i}{Dt} = -\frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j}\left[\mu\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} - \frac{2}{3}\frac{\partial u_k}{\partial x_k}\delta_{ij}\right)\right] + \rho g_i \tag{13}$$

for incompressible flow, Eq. (2) tells us $\frac{\partial u_k}{\partial x_k} = 0$, and together with a constant dynamic viscosity μ so Eq. (13) reduces to:

$$\rho\frac{Du_i}{Dt} = -\frac{\partial p}{\partial x_i} + \mu\frac{\partial^2 u_i}{\partial x_j^2} + \rho g_i \tag{14}$$

This equation represents a balance between inertial forces $\left(\frac{Du_i}{Dt}\right)$, pressure forces $\left(\frac{\partial p}{\partial x_i}\right)$, viscous forces $\left(\mu\frac{\partial^2 u_i}{\partial x_j^2}\right)$ and the body force due to gravity (ρg_i)

1.4 Thermal energy equations/conservation of energy

Schubert et al., 2001 presents various different formulations of the energy equation.

The general form of the energy equation (6.9.1) is:

$$\rho T\frac{Ds}{Dt} = \tau_{ij}\frac{\partial u_i}{\partial x_j} + \frac{\partial}{\partial x_j}\left(k\frac{\partial T}{\partial x_j}\right) + \rho H \tag{15}$$

The terms on the right side of Eq. (15) are, respectively, volumetric heat production rates due to viscous dissipation (internal friction), thermal conduction, and internal heat generation (radiogenic?). Where it is assumed that Fourier's law of heat conduction for an isotropic medium is applicable:

$$q_i = -k\frac{\partial T}{\partial x_i} \tag{16}$$

Where q_i are the components of the heat flux vector. Additionally, the viscous dissipation term can be written as the dissipation function:

$$\Phi = \tau_{ij}\frac{\partial u_i}{\partial x_j} \tag{17}$$

Inserting the definition of our deviatoric stress tensor with $k_B = 0$ (Eq (12)), this becomes

$$\Phi = 2\mu\left\{\frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right) - \frac{1}{3}\left(\frac{\partial u_k}{\partial x_k}\right)\delta_{ij}\right\}^2 \tag{18}$$

For an incompressible and Newtonian fluid ($\frac{\partial u_i}{\partial x_i} = 0$), the dissipation function reduces to

$$\Phi = \frac{\mu}{2}\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right)^2 \tag{19}$$

With various thermodynamic relationships:

$$\alpha = \frac{1}{v}\left(\frac{\partial v}{\partial T}\right)_p \tag{20}$$

$$c_p = T\left(\frac{\partial s}{\partial T}\right)_p \tag{21}$$

$$\left(\frac{\partial s}{\partial p}\right)_T = -\left(\frac{\partial v}{\partial T}\right)_p \tag{22}$$

It can be shown that

$$\frac{Ds}{Dt} = \frac{c_p}{T}\frac{DT}{Dt} - \frac{\alpha}{\rho}\frac{Dp}{Dt} \tag{23}$$

Plugging 23 into Eq. (15) gives

$$\begin{aligned} \rho c_p \frac{DT}{Dt} - \alpha T \frac{DP}{Dt} &= \frac{\partial}{\partial x_i}\left(k\frac{\partial T}{\partial x_i}\right) + \Phi + \rho H \\ \rho c_p \left(\frac{DT}{Dt} - \frac{\alpha T}{\rho c_p}\frac{DP}{Dt}\right) &= \frac{\partial}{\partial x_i}\left(k\frac{\partial T}{\partial x_i}\right) + \Phi + \rho H \end{aligned} \tag{24}$$

2 Schubert et al., 2001 Approximate equations

In this section, Schubert derives the approximate forms of the equations of conservation of energy and momentum appropriate to mantle convection.

2.1 Linearizing the equation of state (density ρ)

From now on, density ρ is considered a function of temperature T and pressure p only, neglecting the density changes associated with mineral phase transitions and changes in chemical composition.
Most of the density change in the mantle is due to hydrostatic compression. The mantle density increases by approximately 65% between the surface and the core-mantle boundary. This is Schubert's justification for assuming a linearized equation of state for mantle density:

$$\begin{aligned} \rho &= \overline{\rho}(\overline{T}, \overline{p}) + \rho' \\ \rho &= \overline{\rho}(\overline{T}, \overline{p}) + \left(\frac{\partial \rho}{\partial p}\right)_T p' + \left(\frac{\partial \rho}{\partial T}\right)_p T' \\ \rho &= \overline{\rho}(\overline{T}, \overline{p}) + \overline{\rho}\overline{\chi}_T p' - \overline{\alpha}\overline{\rho} T' \end{aligned} \tag{25}$$

with

$$\begin{aligned} T &= \overline{T} + T' \\ p &= \overline{p} + p' \end{aligned} \tag{26}$$

Where the overbar ($\overline{\rho}$) refers to a reference state and the primes (ρ') departures from it. The reference state is assumed motionless and steady:

- motionless: $\overline{u_i} = 0$
- steady: $\frac{\partial \overline{u_i}}{\partial t} = 0$
- motionless and steady: $\frac{D\overline{u_i}}{Dt} = 0$

- $\tau_{ij} = \mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} - \frac{2}{3} \frac{\partial u_k}{\partial x_k} \delta_{ij} \right) = 0$ (12)

This reduces the momentum equation ((13), 4) to its static form:

$$0 = -\frac{\partial \overline{p}}{\partial x_i} + \overline{\rho g_i} \tag{27}$$

Integrating equation 27 results in the hydrostatic equation. Further needed for the caluclation of the reference state is the equation for the adiabatic temperature gradient (since the reference state temperature is assumed adiabatic)

$$\left(\frac{\partial T}{\partial x_i} \right)_s = \frac{\alpha T g_i}{c_p} \tag{28}$$

And a specification of the reference state equation of state $\overline{\rho} = \overline{\rho}(\overline{T}, \overline{\rho})$.

2.2 Nondimensionalization

Now, the linearized equation of state (25) will be nondimensionalized with scaling factors consisting of representative/characteristic values presented in table 2. With these, dimensionless variables (denoted with an asterisk *) are introduced by dividing the dimensional variables with their characteristic counterpart.

Our starting point is Eq. (25), together with the thermodynamical definitions of the thermal expansivityand isothermal compressibility, respectively:

$$\alpha = -\frac{1}{\rho} \left(\frac{\partial \rho}{\partial T} \right)_p \tag{29}$$

$$\chi_T = \frac{1}{\rho} \left(\frac{\partial \rho}{\partial p} \right)_T \tag{30}$$

Eq. (25) becomes

$$\begin{aligned} \rho &= \overline{\rho}\,(\overline{T}\,, \overline{p}\,) + \overline{\rho}\,\overline{\chi}_T p' - \overline{\rho}\,\overline{\alpha}\,T' \\ \frac{\rho}{\overline{\rho}} &= 1 + \overline{\chi}_T p' - \overline{\alpha}\,T' \end{aligned} \tag{31}$$

Now, we introduce scaling factors to nondimensionalize the above equation.

- Trivial are the characteristic values χ_{T_r}, α_r, ΔT_r and ρ_r.
- for the velocity scale u_r we introduce the Péclet number, which measures the magnitude of advection relative to diffusion, and assume Pe = 1 (advection and diffusion of same order of magnitude):

$$\begin{aligned} \mathrm{Pe} &= \frac{u_r b}{\alpha_r} = \frac{u_r \rho_r c_{p_r} b}{k_r} = 1 \\ u_r &= \frac{k_r}{\rho_r c_{p_r} b} \end{aligned} \tag{32}$$

- Assuming comparability in magnitude of pressure and viscous forces in the Navier-Stokes equation (with constant dynamic viscosity μ) (Eq. 14):

$$\begin{aligned} \mathcal{O}\left(\frac{\partial p}{\partial x_i} \right) &= \mathcal{O}(\mu \frac{\partial^2 u_i}{\partial x_j^2}) \\ \frac{p_r}{b} &= \mu_r \frac{u_r}{b^2} \\ p_r &= \mu_r \frac{u_r}{b} = \frac{\mu_r k_r}{\rho_r c_{p_r} b^2} \end{aligned} \tag{33}$$

Plugging in $\rho = \rho^* \rho_r$ and similar for the other variables into Eq. (31) gives

$$\begin{aligned}\frac{\rho^*}{\overline{\rho}^*} &= 1 + \overline{\chi}_T^* \chi_{T_r} \frac{p'^* \mu_r k_r}{\rho_r c_{p_r} b^2} - \overline{\alpha}^* \alpha_r T'^* \Delta T_r \\ &= 1 + \overline{\chi}_T^* p'^* \left(\frac{k_r^2 \chi_{T_r}}{\rho_r c_{p_r}^2 b^2} \cdot \frac{\mu_r C_{p_r}}{k_r} \right) - \overline{\alpha}^* T'^* (\alpha_r \Delta T_r) \\ &= 1 + \overline{\chi}_T^* p'^* \mathrm{M}^2 \mathrm{Pr} - \overline{\alpha}^* T'^* \epsilon\end{aligned} \tag{34}$$

where

$$\mathrm{M}^2 = \frac{k_r^2 \chi_{T_r}}{\rho_r c_{p_r}^2 b^2}, \quad \mathrm{Pr} = \frac{\mu_r C_{p_r}}{k_r}, \quad \epsilon = \alpha_r \Delta T_r \tag{35}$$

which are the Mach number (squared, measuring the influence of compressibility), Prandtl number (ratio of momentum diffusivity to thermal diffusivity, or, equivalently the ratio between advection and conduction) and the fractional density change due to temperature variations, respectively. I refer to table 3 for an overview of all dimensionless parameters and characteristic values used in the preceding equations.

Plugging in the values from table 2 gives $Pr \approx 2.5 \times 10^{23}$, which is considered practically infinite, and $M^2 \approx 10^{-33}$ and consequently $M^2 Pr \approx 2 \times 10^{-10} \ll 1$. If $\epsilon \ll 1$ as well, $\frac{\rho^*}{\overline{\rho}^*} \approx 1$ and density perturbations tue to mantle convection are small compared to the reference hydrostatic density of the mantle.
The dimensionless form of the **conservation of mass equation** (Eq. (1)) can be obtained in the same manner: do I have to do this whole derivation as well?

$$\begin{aligned}&\overline{\rho}^* \overline{\chi}_T^* M^2 Pr \frac{\partial p'^*}{\partial t^*} - \overline{\alpha}^* \overline{\rho}^* \epsilon \frac{\partial T'^*}{\partial t^*} \\ &+ \frac{\partial}{\partial x_i^*} \left[\left\{ \overline{\rho}^* + \overline{\rho}^* \overline{\chi}_T^* p'^* M^2 Pr - \overline{\rho}^* \overline{\alpha}^* \epsilon T'^* \right\} u_i^* \right] = 0\end{aligned} \tag{36}$$

The dimensionless and scaled **Navier-Stokes equation** (from Eq. 13 and Eq. 34) is

$$\begin{aligned}&\frac{1}{Pr} \left\{ \overline{\rho}^* + \overline{\rho}^* \overline{\chi}_T^* M^2 Pr\, p'^* - \overline{\rho}^* \overline{\alpha}^* \epsilon T'^* \right\} \frac{D u_i^*}{D t^*} = -\frac{\partial p'^*}{\partial x_i^*} + \overline{g}_i^* \overline{\rho}^* \overline{\chi}_T^* p'^* \frac{D}{\gamma_r} \frac{c_{p,r}}{c_{v,r}} \\ &- \overline{g}_i^* \overline{\rho}^* \overline{\alpha}^* T'^* Ra + \frac{\partial}{\partial x_j^*} \left\{ \mu^* \left(\frac{\partial u_i^*}{\partial x_j^*} + \frac{\partial u_j^*}{\partial x_i^*} - \frac{2}{3} \delta_{ij} \frac{\partial u_k^*}{\partial x_k^*} \right) \right\}\end{aligned} \tag{37}$$

where the dimensionless parameter Ra is the Rayleigh number:

$$Ra \equiv \frac{\alpha_r \, \Delta T_r \, \rho_r^2 \, g_r \, b^3 \, c_{p,r}}{\mu_r \, k_r} \tag{38}$$

The Rayleigh number controls the strenght of the buoyancy force term associated with temperature variations, which is the major driving force of thermal convection.

The scaled and nondimensional **energy equation** (24) is

$$\begin{aligned}&\left\{ \overline{\rho}^* + \overline{\rho}^* \overline{\chi}_T^* M^2 Pr\, p'^* - \overline{\rho}^* \overline{\alpha}^* \epsilon T'^* \right\} \left(\overline{c}_p^* \right) \frac{D}{D t^*} \left(\overline{T}^* + T'^* \right) \\ &\quad - \overline{\alpha}^* \left(\overline{T}^* + T'^* \right) \frac{D p'^*}{D t^*} \frac{\epsilon D}{Ra} - \overline{\alpha}^* \left(\overline{T}^* + T'^* \right) \overline{g}_i^* u_i^* \overline{\rho}^* D \\ &= \frac{\partial}{\partial x_i^*} \left\{ \overline{k}^* \frac{\partial}{\partial x_i^*} \left(\overline{T}^* + T'^* \right) \right\} + \Phi^* \frac{D5}{Ra} \\ &\quad + \left(\overline{\rho}^* + \overline{\rho}^* \overline{\chi}_T^* M^2 Pr\, p'^* - \overline{\rho}^* \overline{\alpha}^* \epsilon^* T'^* \right) H^* \left(\frac{b^2 H_r \rho_r}{k_r \, \Delta T_r} \right)\end{aligned} \tag{39}$$

2.3 Approximations

In the limit of $\epsilon \to 0$ and $M^2 Pr \to 0$, eq. 36 reduces to

$$\frac{\partial}{\partial x_i^*} \left(\overline{\rho}^* u_i^* \right) = 0 \tag{40}$$

Which is known as the **anelastic conservation of mass equation** or the **liquid anelastic approximation** (Jarvis and Mckenzie, 1980). The absence of the $\frac{\partial \rho'}{\partial t}$ term eliminates seismic waves from the set of equations governing mantle convection. This makes sense physically, since seismic activity and mantle convection occur on very different time scales. If, additionally, $\overline{\rho}$ is considered constant with depth, eq. 40 reduces to the incompressible continuity equation:

$$\frac{\partial u_i}{\partial x_i} = 0 \tag{41}$$

This approximation is called the **Boussinesq approximation**.For this approximation to be appropriate, the change in $\overline{\rho}$ with depth has to be investigated. Using the adiabatic compressibility

$$\chi_a = \frac{1}{\rho}\left(\frac{\partial \rho}{\partial p}\right)_s \tag{42}$$

and the hydrostatic equation

$$\frac{\partial \overline{\rho}}{\partial x_i} = \overline{\rho}\,\overline{g_i} \tag{43}$$

gives the density gradient

$$\begin{aligned}\frac{\partial \overline{\rho}}{\partial x_i} &= \left(\frac{\partial \overline{\rho}}{\partial \overline{p}}\right)_s \frac{\partial \overline{p}}{\partial x_i} = \overline{\rho}\,\overline{\chi}_a \frac{\partial P}{\partial x_i} \\ &= \overline{\rho}^{\,2}\chi_a \overline{g}_i\end{aligned} \tag{44}$$

The local density scale height is defined as

$$\begin{aligned}\overline{h}_d &\equiv \left\{\frac{1}{\overline{\rho}}\left|\frac{\partial \overline{\rho}}{\partial x_i}\right|\right\}^{-1} \\ &= (\overline{\rho}\,\overline{\chi}_a \overline{g})^{-1} \\ &= \frac{\overline{\gamma}\,\overline{c}_p}{\overline{\alpha}\,\overline{g}}\end{aligned} \tag{45}$$

where $\gamma = \frac{\alpha}{\rho c_p \chi_a}$ is the Grünseisen parameter. The change in $\overline{\rho}$ with depth is negligible (and the Boussinesq/ incompressible approximation applicable) if $\frac{b}{\overline{h}_d} \ll 1$. Assuming characteristic values, this condition gives

$$\begin{aligned}\frac{b}{\overline{h}_d} &= \frac{1}{\gamma_r}\frac{\alpha_r g_r b}{c_{p_r}} \\ &= \frac{D}{\gamma_r} \ll 1\end{aligned} \tag{46}$$

With D the dissipation number. For the Earth's mantle, $D/\gamma_r \approx 0.5$, and the incompressible mass conservation equation is not very accurate. However, due to its simplicity, it is often adopted into models of mantle convection where the effect of compressibility are minor.

Returning to the (dimensionless and scaled) Navier-Stokes equations, we can see that when

$$\frac{1}{Pr} \to 0 \quad and \quad M^2 Pr \to 0 \tag{47}$$

eq. 37 reduces to its **anelastic (liquid) approximation**:

$$\begin{aligned}0 = &-\frac{\partial p'^*}{\partial x_i^*} + g_i^*\,\overline{\rho}^*\,\overline{\chi}_T^*\,p'^*\,\frac{D}{\gamma_r}\,\frac{c_{p,r}}{c_{v,r}} - g_i^*\,\overline{\rho}^*\,\overline{\alpha}^*\,T'^*\,Ra \\ &+ \frac{\partial}{\partial x_j^*}\left\{\mu^*\left(\frac{\partial u_i^*}{\partial x_j^*} + \frac{\partial u_j^*}{\partial x_i^*} - \frac{2}{3}\,\delta_{ij}\,\frac{\partial u_k^*}{\partial x_k^*}\right)\right\}\end{aligned} \tag{48}$$

and the dimensional form

$$0 = -\frac{\partial p'}{\partial x_i} + \overline{\rho}\,\overline{g}_i\,(\overline{\chi}_T\,p' - \overline{\alpha}\,T') + \frac{\partial}{\partial x_j}\left\{\mu\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} - \frac{2}{3}\,\delta_{ij}\,\frac{\partial u_k}{\partial x_k}\right)\right\} \tag{49}$$

With the additional limit of $D/\gamma_r \to 0$ (together with incompressibility and $\overline{\rho^*} = \overline{\alpha^*} = 1$) the dimensionless Navier-Stokes equations further reduce to the **Boussinesq approximation**:

$$0 = -\frac{\partial p'^*}{\partial x_i^*} - \overline{g}_i^* \, T'^* \, Ra + \frac{\partial}{\partial x_j^*} \left\{ \mu^* \left(\frac{\partial u_i^*}{\partial x_j^*} + \frac{\partial u_j^*}{\partial x_i^*} \right) \right\} \tag{50}$$

and the dimensional form

$$0 = -\frac{\partial p'}{\partial x_i} - \overline{\rho} \, \overline{g}_i \, \overline{\alpha} \, T' + \frac{\partial}{\partial x_j} \left\{ \mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) \right\} \tag{51}$$

Returning to the (nondimensional) heat equation (39) and applying the conditions of $M^2 Pr \to 0$ and $\epsilon \to 0$ for the **anelastic approximation**, this reduces to

$$\begin{aligned} &\overline{\rho}^* \, \overline{c}_p^* \frac{D}{Dt^*} \left(\overline{T}^* + T'^* \right) - \overline{\alpha}^* \left(\overline{T}^* + T'^* \right) \overline{g}_i^* \, u_i^* \, \overline{\rho}^* \, D \\ &= \frac{\partial}{\partial x_i^*} \left\{ \overline{k}^* \frac{\partial}{\partial x_i^*} \left(\overline{T} + T' \right) \right\} + \Phi^* \frac{D}{Ra} + \overline{\rho}^* \, H^* \left(\frac{b^2 H_r \rho_r}{k_r \, \Delta T_r} \right) \end{aligned} \tag{52}$$

Now, using thermodynamic relations in combination with the hydrostatic equation (6.9.17+6.9.18) give

$$\left(\frac{\partial T}{\partial x_i} \right)_s = \frac{\alpha T g_i}{c_p} \tag{53}$$

or, in dimensionless form

$$\left(\frac{\partial \overline{T}^*}{\partial x_i^*} \right)_s = \frac{\overline{\rho}^* \overline{T}^* D \overline{g}_i^*}{\overline{c}_p^*} \tag{54}$$

Where the dissipation number D arises from the nondimensionalization. With this relation, the $\overline{T}^*$ terms in the RHS of equation 52 cancel out and the **anelastic, dimensionless form of the energy equation** becomes

$$\begin{aligned} \overline{\rho}^* \, \overline{c}_p^* \frac{DT'^*}{Dt^*} - T'^* \, \overline{\alpha}^* \, \overline{g}_i^* \, u_i^* \, \overline{\rho}^* \, D = &\frac{\partial}{\partial x_i^*} \left\{ \overline{k}^* \frac{\partial}{\partial x_i^*} \left(\overline{T}^* + T'^* \right) \right\} \\ &+ \Phi^* \frac{D}{Ra} + \overline{\rho}^* \, H^* \left(\frac{b^2 H_r \rho_r}{k_r \, \Delta T_r} \right) \end{aligned} \tag{55}$$

and in dimensional form

$$\overline{\rho} \, \overline{c}_p \frac{DT'}{Dt} - T' \, \overline{\alpha} \, \overline{g}_i \, u_i \, \overline{\rho} = \frac{\partial}{\partial x_i} \left\{ \overline{k} \frac{\partial}{\partial x_i} \left(\overline{T} + T' \right) \right\} + \Phi + \overline{\rho} \, H \tag{56}$$

If $\overline{k}^*$, $\overline{\alpha}^*$, $\overline{c}_p^*$ and $\overline{g}_i^*$ are considered constant, then the terms representing heat conduction along the adiabat become INSERT EQ 6.10.31-32 HERE where eq. 54 was used.

Assuming $D \to 0$, which physically means assuming the representative thermal expansivity α_r to be zero, we arrive at the **Boussinesq energy equation**. INSERT EQ. 6.10.34-48 HERE

3 Appendix 1: tables of constants and reference/scaling values

Table 1: Constants used by Schubert et al., 2001

Constant	Value	Name
k_b	0	Bulk viscosity
α		thermal expansivity
χ_T		isothermal compressibility
c_v		specific heat at constant volume
c_p		specific heat at constant pressure
χ_a		adiabatic compressibility
k		thermal conductivity
H		rate of internal heat production per unit mass
K_a		adiabatic bulk modulus
K_T		isothermal bulk modulus
Φ	$\tau_{ij}\frac{\partial u_i}{\partial x_j}$	dissipation function (eq. 6.9.3)

Table 2: Reference values used by Schubert et al., 2001

Symbol	Value	Description
ΔT_r		Characteristic temperature difference driving thermal convection
χ_{T_r}		Representative isothermal compressibility
α_r		Representative thermal expansivity
u_r	$\frac{k_r}{\rho_r c_{p_r} b}$	Velocity scale (follows from assuming $Pe = 1$)
μ_r	10^{21} Pa s	Representative viscosity
b		Depth of convecting region
p_r	$\frac{\mu_r u_r}{b}$	Scale for dynamic pressure variations
k_r	4 W m^{-1} K^{-1}	Reference value thermal conductivity
c_{p_r}	1 kJ kg^{-1} K^{-1}	Reference specific heat at constant pressure
t_r	$\frac{b}{u_r} = \frac{\rho_r c_{p_r} b^2}{k}$	reference timescale

Table 3: Dimensionless parameters and their values for the Earth's mantle

Parameter	Definition	Value	Description
Pr	$\frac{\mu_r c_{p_r}}{k_r}$	2.5×10^{23}	Prandtl number
M^2	$\frac{k_r^2 \chi_{T_r}}{\rho_r c_{p_r}^2 b^2}$	10^{-33}	Mach number (squared)
ϵ	$\alpha_r \Delta T_r$		Fractional density change due to temperature variations
Pe	$\frac{u_r b}{\alpha_r} = \frac{u_r \rho_r c_{p_r} b}{k_r}$		Péclet number for heat transfer (advection/diffusion)